

Riparian plant guilds of ephemeral, intermittent and perennial rivers

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SUMMARY

1. Plant functional types (or guilds) increasingly are being used to predict vegetation response to global changes. Continued human population growth coupled with projected warmer and drier climate will alter the hydrologic regimes of many arid-zone rivers, including intermittent rivers. We aimed to identify (i) woody plant guilds associated with distinct stream types of an arid region and (ii) plant traits indicating adaptation to the selective pressures of water availability and fluvial disturbance.

2. We used hierarchical clustering to identify 11 plant guilds from floodplains, terraces and uplands of eight Arizona rivers that vary in surface flow permanence, depth to ground water and intensity of fluvial disturbance.

3. Six guilds were riparian pioneers with small, wind-dispersed seeds, three guilds were late-seral, shade-tolerant riparian taxa with large animal-dispersed seeds, and two guilds were composed of desert xerophytes. Within the riparian pioneer and seral groups, guilds varied in water acquisition and productivity traits including wood density and rooting depth.

4. The community-weighted traits varied or covaried with water availability and fluvial disturbance. Root: shoot ratio, canopy height and leaf area were influenced strongly by water availability, with the latter two showing a nonlinear response to changes in water table depth. Leaf length increased, and wood density decreased, as sites become wetter and more fluvially disturbed. Community-weighted seed mass, seed dispersal and spinescence varied most strongly with elevation above thalweg (an indicator of decreasing fluvial disturbance).

5. These analyses will enable prediction of changes in the relative abundance of plant types and plant traits in response to changes in stream flow regimes, such as shifts towards greater intermittency. The distribution patterns of guilds among riparian habitat types emphasise the importance of focusing conservation efforts not only on the limited number of perennial rivers remaining in arid regions, but also on intermittent and ephemeral rivers with shallow water tables.

Keywords: desert stream, plant functional type, riparian vegetation, trait, wood density

Introduction

Anthropogenic changes to stream hydrology have substantially altered the distribution, abundance and species composition of riparian plant communities (Patten, 1998; Tockner & Stanford, 2002; Stella *et al.*, 2012). Increasing global pressures on freshwater resources, coupled with climate change, will drive further changes in stream surface and subsurface flows (Palmer *et al.*, 2009).

Alterations are likely to be greatest in highly populated dryland regions (Döll, Fiedler & Zhang, 2009).

Water is the primary limiting factor for vascular plants in dryland regions. In some desert rivers, stream flows are increasing owing to effluent discharge, agricultural return flow and urbanisation, but for many others, flows are decreasing owing to groundwater pumping, diversion of surface water and climate-linked reductions in stream and groundwater recharge (Serrat-Capdevila

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et al., 2007; Jobbágy *et al.*, 2011). Many desert rivers have spatially varying flows over their length, and the perennial segments (already rare in arid landscapes) increasingly are becoming intermittent (Larned *et al.*, 2010; Steward *et al.*, 2012; Seager *et al.*, 2013). At the same time, depth to ground water is increasing in the alluvial aquifers that sustain riparian phreatophytes, due to reduced recharge and accelerated groundwater pumping (Castle *et al.*, 2014). Flood patterns also are changing. Although some dams are being removed or some elements of flow regime restored via minimum flow standards and/or conservation floods, people continue to build and enlarge existing flow-regulating dams and reservoir systems to capture flood water. Unregulated rivers, in contrast, are experiencing more intense floods (as well as reduced low flows) as the global climate changes (Dominguez, Cañon & Valdes, 2010; Dallas & Rivers-Moore, 2014).

Stream and riparian management is informed by knowing the hydrologic ranges of conservation targets and their expected shifts in distribution following forecasted changes in flow regimes (Eamus & Froend, 2006). Researchers in dryland regions worldwide have described how species composition and plant growth forms change as hydrologic thresholds of flow permanence and groundwater depth are crossed and as flood patterns deviate from historic norms (e.g. Liu *et al.*, 2005; Merritt & Poff, 2010; Gonzalez-Sanchis, Comin & Muller, 2012). A need remains, however, for a more globally transferable approach that goes beyond taxonomy and beyond simple life-form or region-of-origin classification schemes.

Plant functional traits and types increasingly are being used to predict vegetation responses to environmental changes at local and global scales (Díaz & Cabido, 1997; Gitay & Noble, 1997). For conservation and management, plant functional types need to be tailored for the particular stressors and limiting factors of a given ecoregion (Grime, 2001; Merritt *et al.*, 2010; Eallonardo *et al.*, 2013). In dryland regions, functional types have yet to be comprehensively described for riparian plants. Some researchers have delineated functional groups that have similar responses to soil wetting and drying patterns (Brock & Casanova, 1997; Casanova, 2011). Other studies have used 'habitat preference guilds' such as wetland indicator groups that synthesise the combination of traits that adapt plants to different levels of water availability (Stromberg *et al.*, 2005; Goebel, Pregitzer & Palik, 2006; Laidig *et al.*, 2010). Such studies help us understand how non-phylogenetic groups respond to hydrologic changes, such as shifts from perennial to intermittent

flows, but the need to capture the causal mechanisms underlying habitat preferences and tolerances remains (Merritt *et al.*, 2010; Campbell, Johns & Nielsen, 2014). Traits that reflect productivity and growth rate, regeneration processes, storage and defence all are important when identifying suites of species adapted to riparian conditions (Hunt *et al.*, 2004; Aguiar *et al.*, 2013; Verberk, van Noordwijk & Hildrew, 2013).

The objectives of this study were to (i) identify riparian plant functional types associated with distinct hydrologic habitat types encompassing gradients of water availability and fluvial disturbance and (ii) determine how selected functional traits vary along the same two gradients. The study system was the riparian zones of ephemeral, intermittent and perennial reaches of arid south-western U.S.A. With regard to plant traits, we hypothesised that (i) water acquisition and productivity are strongly related to depth to water table and perenniality of flows, with smaller canopies, smaller leaves, deeper roots, higher root: shoot ratios, higher water use efficiency (wood density) and more prevalent spinescence in areas where water is scarce (deep water table, less frequent flow) and (ii) reproductive traits are strongly related to fluvial disturbance, with small, wind- and water-dispersed seeds more prevalent in areas of higher fluvial disturbance than in more stable habitats, where animal-dispersed seeds would prevail.

Methods

Experimental design and study sites

Eight undammed drainages (four rivers and four streams) in the Sonoran Desert of central and southern Arizona, U.S.A., served as study areas. Variability in physical conditions derived from three sources: between rivers ($n = 8$), longitudinal location within a river ($n = 24$ river sites) and lateral position within a river site ($n = 204$ riparian plots). Between and within rivers, there were differences in flow permanence (Fig. 1). We defined perennial, intermittent and ephemeral stream reaches, respectively, as those in which surface flow was present 100% of the time, 26–95% of the time and $\leq 25\%$ of the time. In the sand-bed streams of desert basins, surface flows and depth to water table usually covary (Lite & Stromberg, 2005; Merritt & Bateman, 2012). Perennial surface flows are sustained by inflowing ground water and are associated with shallow and stable water tables. Intermittent flows are characteristic of hydrologically losing reaches with deeper and more fluctuating water tables. Ephemeral streams normally

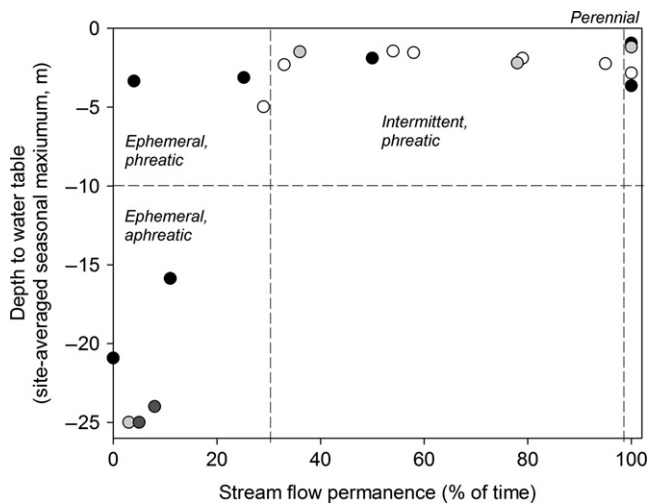


Fig. 1 Depth to water table in the river floodplain zone versus the frequency with which surface flow is present, for reaches of four rivers in the Sonoran Desert, Arizona. Each river is represented by a different fill shade.

are nonphreatic with transient water tables developing after major storm run-off (Rassam *et al.*, 2006). Our study included ephemeral nonphreatic streams as well as the less studied ephemeral phreatic streams that have more permanent, shallow water tables (Shaw & Cooper, 2008).

The four study rivers (Hassayampa, Lower Santa Cruz, Lower Cienega Creek and Lower San Pedro) drain large catchments and are subject to frequent scouring floods (Stromberg, 2007; Stromberg, Hazelton & White, 2009a; Stromberg *et al.*, 2009b). Perennial reaches co-occur with intermittent to ephemeral reaches over scales of metres to kilometres owing to local differences in hydrogeology. There is considerable topographical variation within the river floodplains, owing to sedimentation during depositional floods and channel incision. Further, the river beds incised a century ago, creating sharp lateral discontinuities between the floodplain (post-entrenchment surfaces) and the unflooded terraces (pre-entrenchment surfaces); the terraces are several metres above the floodplains and support phreatophytes (Webb & Leake, 2006). The four study streams have ephemeral flow and are locally referred to as desert washes. Two of the four (Black Gap Wash and Saucedo Wash) are near Gila Bend and two (both unnamed) are south of Tucson.

One to seven study sites (i.e. a river segment 100 m long) were established per river or stream. Study sites range in elevation from 258 to 1010 m (site elevations are metres above mean sea level). Long-term (30 year) mean annual precipitation (<http://www.wrcc.dri.edu/>)

ranges among study areas from 10 to 35 cm; mean annual temperature ranges from 18 to 22 °C. Vegetation in the study area is Sonoran Desert Scrub at lower elevations and degraded desert grassland at higher elevations (Brown, 1994).

Vegetation sampling and trait measurement

At each site, two cross-riparian transect lines were established, separated by a river distance of 100 m. The transect lines spanned the active channel, floodplain and, if present, secondary channels and terraces. Plots along each transect were sampled in stratified random fashion (two-stage sampling) within vegetation zones (patch types). The patch types were delineated based on major differences in vegetation physiognomy or species composition. One to two 100-m² plots (5 × 20 m; long axis parallel to stream) were sampled per zone per transect for a total of 204 riparian plots: 77 at San Pedro River, 66 at Cienega Creek, 39 at Hassayampa River, 10 at Santa Cruz River and three at each desert wash. Twelve plots were sampled in Sonoran Desert Scrub.

We examined woody vegetation only; herbaceous species were excluded because of their high temporal variability. In each plot, diameter of woody plant stems was measured to obtain density and basal area per species. Canopy cover was measured using a densiometer. Plants were identified to species using Kearney & Peebles (1960) and more recent taxonomic treatments. A modified importance value was calculated for each species, per plot, by averaging the species' relative stem density, relative basal area and relative canopy cover (Curtis, 1959).

Functional types can be classified with regard to *effect* (suites of species with similar ecosystem functions) or *response* (species that share morphological and physiological traits that regulate resource use and allocation). In our study, response trait data were obtained or measured for 32 of the 48 woody species found in study plots; the 16 species without trait data were rare components of the community. Leaf area, leaf length, canopy height, rooting depth, root: shoot (depth/height) ratio, wood density, seed mass, seed dispersal morphology and presence of spines or thorns (hereafter, spinescence) were selected as traits reflecting adaptations to water availability, fluvial disturbance and defence. For leaf area, samples of three mature leaves were randomly collected from three plants per species and measured with a LICOR leaf area meter (or, for very small or scale-like leaves, with a ruler). For plants with compound leaves, values are reported for the smallest functional leaf unit,

the leaflet (Givnish, 1987). For *Tamarix aphylla*, a linear segment of overlapping scale-like leaves was considered as the leaf unit. For canopy height, the typical maximum value for the species was recorded using local floras and regional databases (SEINet, 2013). For rooting depth, we used the maximum depth reported in the literature (DePree & Ludwig, 1978; Fonteyn & Mahall, 1981; Stromberg, 2013; J. Shaw, pers. comm., February 2015). Wood density data were obtained from literature searches and unpublished measurements (Hacke *et al.*, 2009; T. Kolb, pers. comm., April 2015). For seed mass, samples from three individuals per species were collected and weighed or values were obtained from the Seed Information Database (Stromberg & Boudell, 2013; Royal Botanic Gardens Kew, 2015). For seed dispersal morphology, two classes were used: wind-dispersed (as indicated by presence of wings or plumes that aid in aerial seed transport) or internally or externally animal-dispersed (fleshy fruit or large nutritious seeds, burrs). For spinescence, species were assigned a score of 0 if spines and thorns were absent and 1 if spines or thorns were present.

Stream hydrology

We obtained four measures of stream hydrology. Two described water resource availability: flow permanence (site level) and depth to water table (plot level); and two described fluvial disturbance: catchment size (site level index of flood intensity and frequency) and elevation above thalweg (plot level index of flood intensity and frequency). Two-year data sets on monthly values of surface flow presence-absence and depth to the water table were obtained either by collecting field data, obtaining information from land managers (e.g. Pima Association of Governments) or accessing data from USGS (<http://waterdata.usgs.gov>) or Arizona Department of Water Resources Groundwater Site Inventory. The flow permanence field data were obtained either from monthly site visits or from downloads from electrical resistance sensors with exposed leads placed in the streambed (Blasch *et al.*, 2002). The data sets were used to generate a value indicating the percentage of time the site had surface flow (i.e. the hydroperiod). The water table data were obtained from monitoring wells located near study transects, and the well sites and study plots were surveyed for elevation of the ground surface above the channel thalweg using a stadia rod and transit. The slope of the water table at each site (assumed to be linear) was estimated using measurements from the monitoring wells in conjunction with the surface stream level. These data collectively were used to estimate, per plot,

the mean annual maximum depth to ground water (which typically occurred in June). Catchment size was obtained from USGS gaging stations or from topographical maps and ranged among rivers from 2 km² (unnamed washes) to 8018 km² (San Pedro River; station 09472000).

Habitat types

We subjectively delineated 10 riparian habitat types. One type was nonphreatic washes with ephemeral flow and a small catchment. Nine types occurred along the large rivers and were based on permutations of (i) surface flow type (perennial, intermittent, ephemeral), (ii) depth to ground water (phreatic with water table shallower than 10 m, nonphreatic) and (iii) fluvial disturbance (frequently flooded surfaces <2 m above the thalweg, infrequently flooded surfaces 2–5.5 m above the thalweg, unflooded stream terraces 5.5–10 m above the thalweg). An 11th habitat type, Sonoran Desert upland, was non-riparian.

Statistical analysis

Hierarchical group-average clustering was used to identify plant functional types using a Gower dissimilarity matrix of traits. The data matrix contained 32 woody species and nine traits. Gower's coefficient is preferred to Euclidean distance measures as Gower's coefficient can elegantly combine different types of descriptors (continuous, binary and ordinal) and process each one according to its own mathematical properties (Legendre & Legendre, 1998). For variables with non-normal distributions (seed mass and leaf area), raw values were natural log-transformed before generating the Gower dissimilarity matrix to more closely comply with assumptions of statistical tests. We tested for significant differences between groupings using similarity profile permutation tests (SIMPROF). SIMPROF tests for significant evidence of genuine clusters in samples that are unstructured *a priori* (Clarke & Gorley, 2015). We calculated the relative abundance (and number) of each plant functional type within each of the 11 subjectively defined habitat types. To test for differences in the hydrologic characteristics of sites associated with each plant functional type, we used Kruskal–Wallis ANOVA and *post hoc* Dunn's tests (Bonferroni adjusted) using data sets of plots with at least 50% relative abundance of each type.

We used linear regression to examine how community-weighted traits of riparian plants vary along hydro-

logic continua. Community-weighted values can be based on mean values for traits or on maxima (Dubuis *et al.*, 2013); we used mean values for all traits except rooting depth and canopy height for which maxima were used. To generate the community-weighted values, each species' fractional importance value in the plot was multiplied by its trait value and these values were summed across all species per plot. There were five plots for which the summed importance value of species without trait data exceeded 20%, and these plots were deleted from analysis. The four variables describing the hydrologic continua (flow permanence, annual maximum depth to water table per plot, catchment size, elevation of plot above channel bed) were natural log-transformed to comply with the assumptions of linear regression analyses. We used adjusted r^2 and Akaike's information criterion (penalised for the number of parameters in the model) to evaluate alternative models and select the best set of independent variables for each community-weighted trait. Elevation above channel bed frequently covaries with many physical properties (Auble, Friedman & Scott, 1994) and was correlated with depth to water table in our study ($r = 0.43$, $P < 0.001$), as were flow permanence and depth to water table ($r = -0.65$, $P < 0.001$).

To test our specific hypothesis, we produced a principle coordinates analysis (PCOa) based on a Gower dissimilarity matrix for water acquisition traits (leaf area, leaf length, canopy height, rooting depth and wood density) and for reproductive traits (seed mass and seed dispersal morphology). Principal coordinates analysis is similar to principal components analysis, but rather than generating linear combinations of the original variables, PCOa creates coordinates using the original variables mediated through a variety of distance matrices, in our case the Gower dissimilarity index (Legendre & Legendre, 1998). We then regressed the four hydrologic variables against the first axis of each PCOa. Regression models were assessed based on their adjusted r^2 and P -values; only significant hydrologic variables were retained in the final models. Final models, depending on dependent variable, had two to four independent variables. We present results for the first PCOa axis from the separate ordinations for each trait category (water acquisition and reproductive).

To focus specifically on ground water, we calculated community-weighted trait means for six depth-to-water-table zones and used Kruskal–Wallis ANOVA and *post hoc* Dunn's tests (Bonferroni adjusted) to test pairwise differences between zones. Regression analyses, Kruskal–Wallis ANOVA and Dunn's tests were per-

formed in R (R Core Team, 2015), and cluster, PCOa and SIMPER analyses were performed using PRIMER-E version 7 (Clarke & Gorley, 2015).

Results

Plant guilds

We delineated a total of 11 guilds (Table 1). SIMPROF tests revealed seven significantly different clusters among the 32 woody plant species. We further subdivided a large cluster that had high internal dissimilarity into four smaller clusters, producing 10 guilds (Fig. 2). The 11th guild consisted of stem succulents, for which little information was available on wood density or rooting depth.

Guilds P1 through P6 are riparian pioneers with small, wind-dispersed seeds (Fig. 2). Guilds S3 through S5 are mid-seral (hereafter, seral), shade-tolerant riparian species with large seeds adapted for animal dispersal. The remaining two guilds (D5 and D6) are desert affiliated. Within the pioneer (P) and seral (S) riparian groups, guilds are distinguished by water acquisition traits including wood density, leaf size, canopy height and rooting depth and by the defence trait of spinescence. We assigned the numeric component of the guild code based on increasing drought tolerance as inferred from wood density; the seral guilds begin at S3 because no guild in this group has mean wood density lower than 0.50 g cm^3 .

The largest guild has eight species; two guilds are singlets. *Prosopis velutina* is the sole member of guild S5, distinguished by small leaves but very deep roots, tall canopy, relatively dense wood, spines and large animal-dispersed seeds. The other singlet, *Atriplex canescens*, would likely group with other species if sites with higher salinity were sampled.

Guild distribution

There were significant differences in stream flow permanence, depth to water table and elevation among sites dominated by different guilds (Fig. 3). Stream flow was more permanent at sites dominated by hydriparian guilds (P1 and P2) than at sites dominated by xeriparian guilds (P5, S4 and S5) (Kruskal–Wallis $H = 42.1$, $P < 0.001$). Depth to water table was significantly greater for shallower-rooted guild S4 (median of 20.8 m) than for deeper-rooted S5 (6.2 m). Among pioneer types, depth to water table was significantly greater for P5 (2.9 m) than for P1 (1.7 m) and P2 (1.5 m) (Kruskal–Wal-

Table 1 Distribution of woody plant species among guilds, based on hierarchical clustering

Guild code	Guild name	Dispersal mode	Seed mass (mg)	Leaf area (cm ²)	Wood density (g cm ⁻³)	Height (m)	Rooting depth (m)	Spinescence	Species
P1	Hydroriparian pioneer tree	w	0.3 ± 0.3	33 ± 21	0.46 ± 0.01	30 ± 0	2 ± 1	n	<i>Populus fremontii</i> (Salicaceae) <i>Salix gooddingii</i> (Salicaceae)
P2	Hydroriparian pioneer shrub	w	0.1 ± 0.0	11 ± 7	0.51 ± 0.05	6 ± 5	2 ± 1	n	<i>Baccharis salicifolia</i> (Asteraceae) <i>Pluchea sericea</i> (Asteraceae) <i>Salix bonplandiana</i> (Salicaceae) <i>Salix exigua</i> (Salicaceae)
P3	Mesoriarian pioneer tree	w	28 ± 12	11 ± 5	0.55 ± 0.18	19 ± 1	3 ± 1	n	<i>Acer negundo</i> (Aceraceae) <i>Fraxinus velutina</i> (Oleaceae)
P4	Xeroriarian pioneer tree, deep-rooted	w	4 ± 4	3 ± 3	0.60 ± 0.00	13 ± 6	13 ± 6	n	<i>Chilopsis linearis</i> (Bignoniaceae) <i>Tamarix aphylla</i> (Tamaricaceae)
P5	Xeroriarian pioneer shrub	w	3 ± 4	1 ± 1	0.76 ± 0.17	3 ± 2	5 ± 4	n	<i>Ambrosia monogyra</i> (Asteraceae) <i>Ambrosia salsola</i> (Asteraceae) <i>Baccharis sarothroides</i> (Asteraceae) <i>Ericameria nauseosa</i> (Asteraceae), <i>Gutierrezia sarothrae</i> (Asteraceae) <i>Tamarix chinensis</i> (Tamaricaceae) <i>Atriplex canescens</i>
P6	Xeroriarian pioneer shrub, deep-rooted	w	8.6	0.8	0.80	2	12	y	
S3	Mesoriarian, late-seral tree	a	199 ± 265	14 ± 7	0.54 ± 0.08	12 ± 5	3 ± 2	n	<i>Celtis reticulata</i> (Cannabaceae) <i>Juglans major</i> (Juglandaceae) <i>Sambucus cerulea</i> (Adoxaceae) <i>Acacia greggii</i> (Fabaceae) <i>Olneya tesota</i> (Fabaceae)
S4	Xeroriarian shrub or small tree, spinescent	a	85 ± 69	0.5 ± 0.4	0.66 ± 0.12	6 ± 4	3 ± 13	y	<i>Parkinsonia florida</i> (Fabaceae) <i>Celtis pallida</i> (Cannabaceae) <i>Lycium andersonii</i> (Solanaceae) <i>Lycium pallidum</i> (Solanaceae) <i>Rhus microphylla</i> (Anacardiaceae) <i>Ziziphus obtusifolia</i> (Rhamnaceae) <i>Prosopis velutina</i> (Fabaceae)
S5	Xeroriarian tree, deep-rooted, spinescent	a	117	0.3	0.77	15	12	y	
D5	Desert shrub	a	8 ± 4	0.7 ± 0.7	1.03 ± 0.16	1 ± 1	2 ± 3	n	<i>Ambrosia dumosa</i> (Asteraceae) <i>Ambrosia deltoidea</i> (Asteraceae) <i>Larrea tridentata</i> (Zygophyllaceae)
D6	Stem succulent	a	12 ± 7	0 ± 0	no data	1 ± 1	no data	y	<i>Cylindropuntia spinosior</i> (Cactaceae) <i>Ferocactus wislizeni</i> (Cactaceae) <i>Opuntia engelmannii</i> (Cactaceae)

The species are from floodplains or terraces (and adjacent upland) of low-elevation rivers and streams in the Sonoran Desert, Arizona. Letters within the alphanumeric guild code differentiate between riparian pioneer (P), riparian seral/shade-tolerant (S) and desert (d) types, and between types that are more (high numbers) and less (low numbers) drought tolerant. (w = wind- or water-dispersed; a = animal-dispersed; n = no, y = yes).

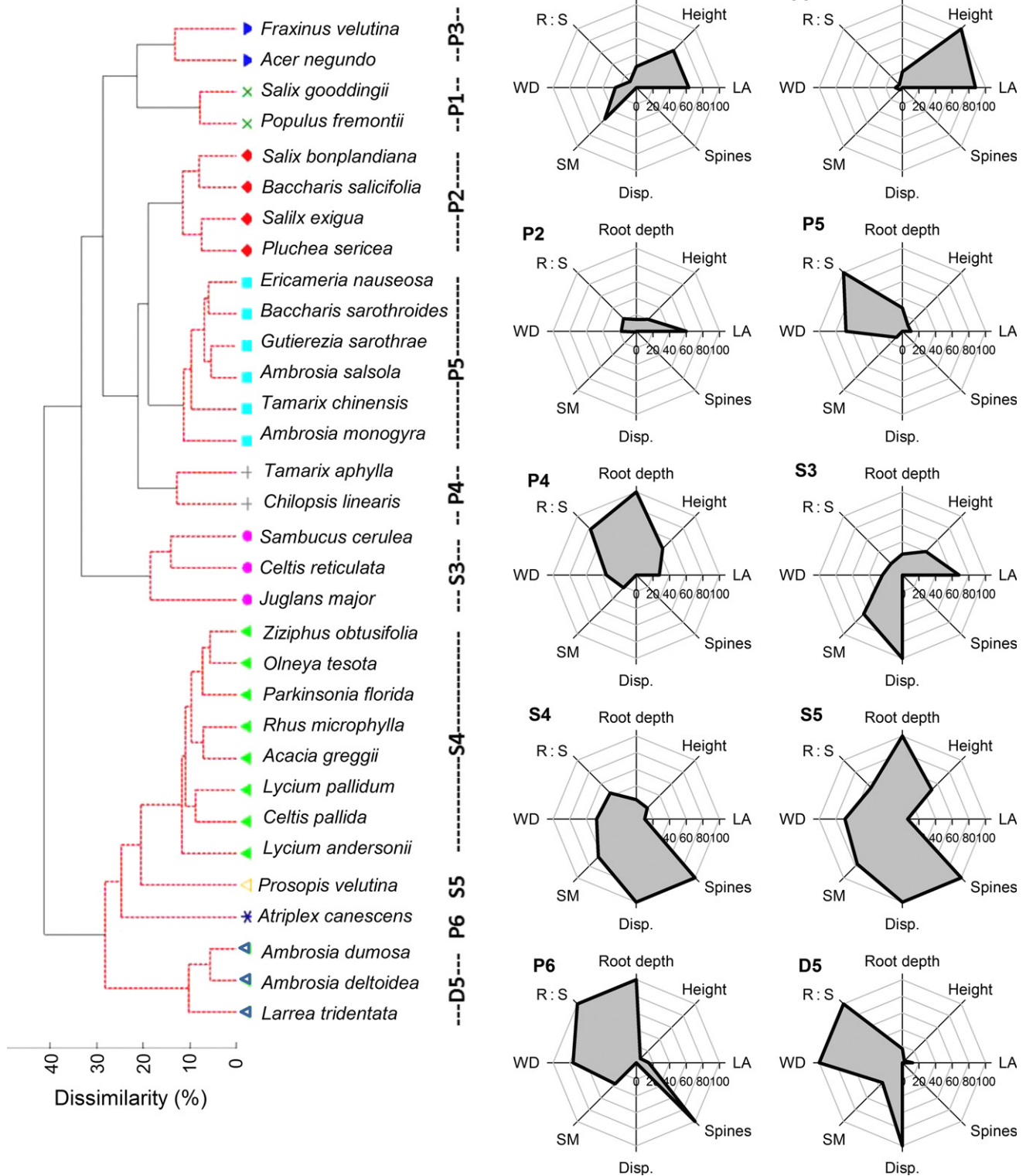


Fig. 2 Dendrogram showing clustering of plant guilds for riparian and desert plants of the Sonoran Desert, Arizona (Table 1). Solid lines on the dendrogram indicated significantly different clusters based on SIMPROF tests. On the right, radar plots show mean trait values, by guild. Trait values are standardised to a maximum of 100. (Height, canopy height; LA, leaf area; Spines, spinescence; Disp., dispersal; SM, seed mass; WD, wood density and R : S, root-to-shoot ratio). High values for dispersal indicate animal (versus wind) dispersal; high values for spinescence indicate presence of spines.

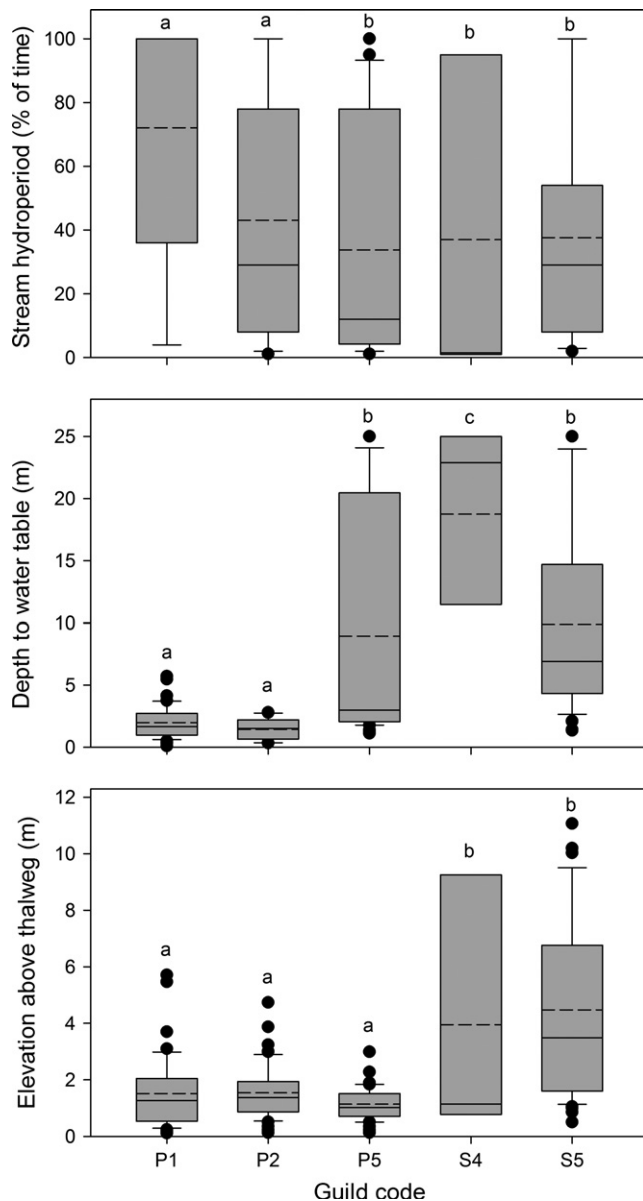


Fig. 3 Box plots indicating the range of hydrologic conditions associated with each of five riparian plant guilds (Table 1). The median is indicated by a solid line and the mean by a dotted line, the box boundaries are the 25th and 75th percentiles, and the whiskers are outliers or extremes. Values are based on sampled plots in which a guild had a relative importance of at least 50%. Guilds with different lower case letters differ significantly in median values at $P < 0.05$, based on Kruskal-Wallis ANOVA on ranks and Dunn's multiple comparison tests.

lis $H = 85.7$, $P < 0.001$). Compared to pioneer guilds, seral guilds were on surfaces that were significantly higher above the channel bed, such as river terraces and high floodplains.

Each riparian habitat type supported multiple guilds, although the dominant guild type varied (Fig. 4).

Hydroriparian guilds P1 and P2 were abundant along perennial rivers with shallow water tables and gave way to xeroriparian pioneers such as *Tamarix chinensis* (P5) as water tables deepened and to seral taxa such as *Celtis reticulata* (S3), *Ziziphus obtusifolia* (S4) and *P. velutina* (S5) as fluvial disturbance decreased. Notably, P1 were present on the intermittent phreatic and ephemeral phreatic stream types but absent from nonphreatic ephemeral streams. The greatest average number of guilds occurred under the combination of wet conditions (perennial rivers with shallow water tables) and intermediate disturbance (high floodplains). Within areas with shallow water tables, tall (30 m) *Populus/Salix* forests (P1) coexisted with shrubby guilds, contributing to high variance in height in wet areas (Fig. 5, upper right). Within areas with deep water tables, deep-rooted xeroriparian trees (e.g. *Chilopsis linearis*; P4) co-occurred with more shallowly rooted xeroriparian shrubs (e.g. *Ambrosia monogyra*; P5) (Fig. 5, lower left).

Community-weighted traits

Each community-weighted trait varied or covaried with water availability and fluvial disturbance in the regression models (Table 2). Three community-weighted traits – root:shoot ratio, canopy height and leaf area – were influenced strongly by water availability. The variability in root:shoot ratio, which ranged among plots from 0.07 to 3.3, was explained primarily by depth to water table. Canopy height increased sharply as water tables became shallower and, to a lesser extent, as elevation above thalweg increased. Patterns for leaf area indicated broad-leaved riparian plants to be most abundant in areas with shallow ground water, high flow permanence and high fluvial disturbance. Canopy height and leaf area declined abruptly as water tables dropped below c. 10 m for the former and c. 6 m for the latter (Fig. 5).

Three community-weighted traits covaried with water availability and fluvial disturbance (Table 2). Wood density increased as sites became drier (lower flow permanence and deeper ground water) and less fluvially disturbed (higher elevations above thalweg). Leaves were longer where water tables were deep and where fluvial disturbance was high (low-elevation plots and large-catchment streams). Rooting depth, surprisingly, was associated with large catchments and infrequent flow.

Community-weighted seed mass and seed dispersal morphology, as expected, varied most strongly with elevation above thalweg pointing to the primary role of fluvial disturbance as an influence on these regenerative

Fig. 4 Relative abundance, as indicated by importance value, of woody plant guilds among riparian habitat types of the Sonoran Desert, Arizona. Some guilds (Table 1) are combined for presentation purposes. Eph., ephemeral; Eph-ph., ephemeral phreatic; Int., intermittent; Per., perennial. Numbers above the bars are the mean (+SE) number of guilds per habitat type.

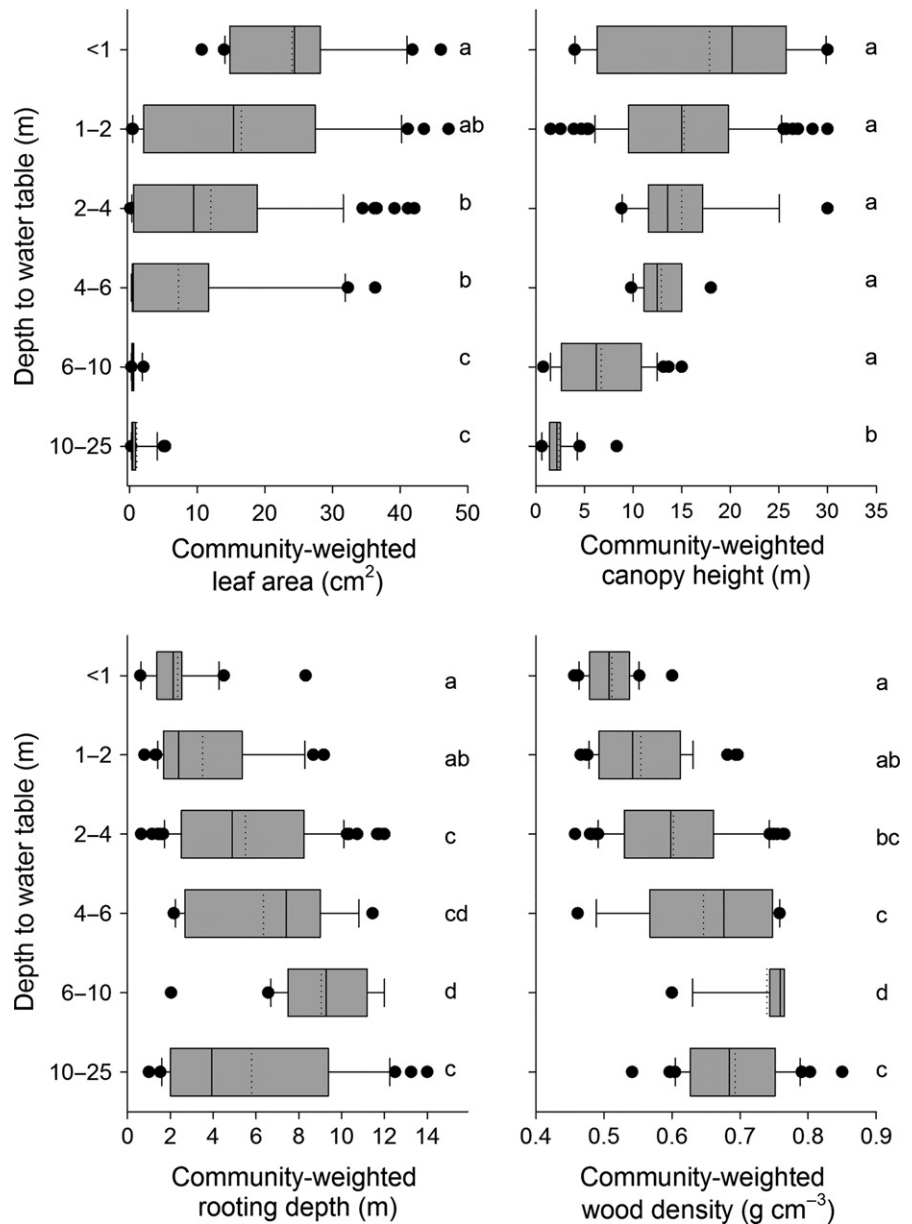
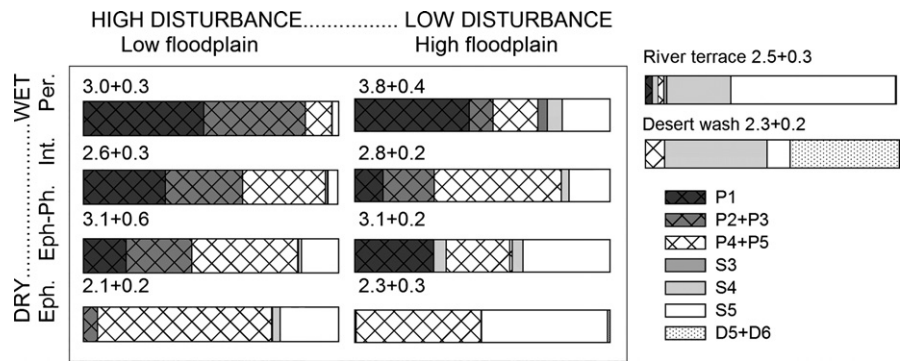


Fig. 5 Box plots of community-weighted leaf area, canopy height, rooting depth and wood density, by depth-to-water-table zone. The median is indicated by a solid line and the mean by a dotted line, the box boundaries are the 25th and 75th percentiles, and the whiskers are outliers or extremes. Zones with different lower case letters differ significantly at $P < 0.05$, based on Kruskal–Wallis ANOVA on ranks and *post hoc* Bonferroni tests (d.f. = 183 or 186).

traits (Table 2). Seed dispersal morphology was also related to catchment size, with wind dispersal prevalent along the large-catchment, high-disturbance rivers. Seed

mass and morphology also were influenced by ground water, with large seeds and animal-mediated dispersal becoming increasingly common on higher (and drier)

Table 2 Regression models relating community-weighted trait variables to abiotic variables

	Water resource variables		Fluvial disturbance variables		Adjusted r^2	F	P -value
	Flow permanence	Depth to water table	Catchment size	Elevation above thalweg			
Strongly water-influenced							
Root: shoot		12.0***	2.6*		0.42	72.5	<0.001
Canopy height		−9.1***		3.1**	0.31	44.9	<0.001
Leaf area	3.0**	−5.8***	−2.9**	−4.4***	0.41	35.4	<0.001
Co-influenced							
Leaf length		−7.1***	4.6***	−8.4***	0.43	50.1	<0.001
Rooting depth	−3.4***		4.1***		0.27	25.7	<0.001
Wood density	−2.9**	4.5***		8.3***	0.44	52.0	<0.001
Strongly disturbance-influenced							
Seed mass		4.4***		9.4***	0.36	57.2	<0.001
Dispersal		5.1***	−2.9**	10.7***	0.44	52.7	<0.001
Spinescence		4.2***		11.2***	0.43	75.7	<0.001
WaterPCOa1	2.7**	−5.8***		−7.0***	0.44	53.9	<0.001
RegenPCOa1		5.1***		10.3***	0.41	70.0	<0.001

Values shown are t -values (tests of significant slope for each relationship; *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$) and the adjusted r^2 , F value and P -value for each model. WaterPCoA1 is axis 1 of the water acquisition PCOA, which accounts for 77% of the variation in leaf area (Spearman rank correlation $r = 0.52$), leaf length ($r = 0.37$), rooting depth ($r = -0.37$), plant height ($r = 0.23$) and wood density ($r = -0.37$). RegenPCoA1 is axis 1 of the regenerative PCOA, which explains 99% of the variation in seed mass ($r = 0.59$) and dispersal mode ($r = 0.47$).

surfaces (Fig. 6). The single defence trait, community-weighted spinescence, increased most strongly with elevation above thalweg and to a lesser extent with increasing depth to ground water.

Analysis of the bundled traits provided partial support for our hypotheses (Table 2). Elevation above thalweg and depth to water table were significantly and positively related to the PCOA axis representing regenerative traits. Two water availability variables, flow permanence and depth to water table, together with the fluvial disturbance variable elevation above thalweg, were significantly related to the PCOA axis representing water acquisition traits, the former two variables showing positive relationships, and the latter negative.

Discussion

Our objectives were to identify plant functional types associated with discrete dryland riparian habitats and to quantify patterns of change in functional plant traits along gradients of water availability and fluvial disturbance. We derived nine distinct riparian guilds (and two upland guilds) from nine quantitative functional traits that vary in predictable ways along the gradients. The distribution of each guild provides insight into its potential habitat, enabling prediction of shifts in guild composition given changing flow regimes, such as increasing occurrence of flow intermittence. Further, the linkages we identified between riparian habitats, guilds and traits will inform conservation prioritisation efforts.

Our results indicate that depth to the water table and fluvial disturbance and, to a lesser but significant extent, degree of flow permanence, govern guild structure in dryland riparian ecosystems. Perennial rivers with shallow water tables underlying the floodplain have been the focus of conservation efforts as they sustain dense and productive riparian forests and high-quality wildlife habitat. Perennial reaches in our study had the greatest abundance of the hydri-rarian guilds, but these same guilds occurred in intermittent as well as ephemeral phreatic reaches, albeit in lower abundance. These results signal the importance of conserving shallow-groundwater stream systems, even where surface flows cease and channel beds dry.

Fluvial disturbance traits and pioneer and seral guilds

Disturbance selects for those species that can complete their life cycle between disturbance events or that can resist or be resilient to damage from disturbance events. Floods in desert rivers exert their greatest influence by scouring, eroding and depositing sediments as well as rewetting the soil column and recharging shallow aquifers. Pioneer species in frequently flood-scoured habitat have many strategies to cope with fluvial disturbance but typically share some combination of traits including short lifespan, rapid growth to reproductive maturity, small seed size, water or wind dispersal, stem flexibility and capacity for vegetative spread and resprouting

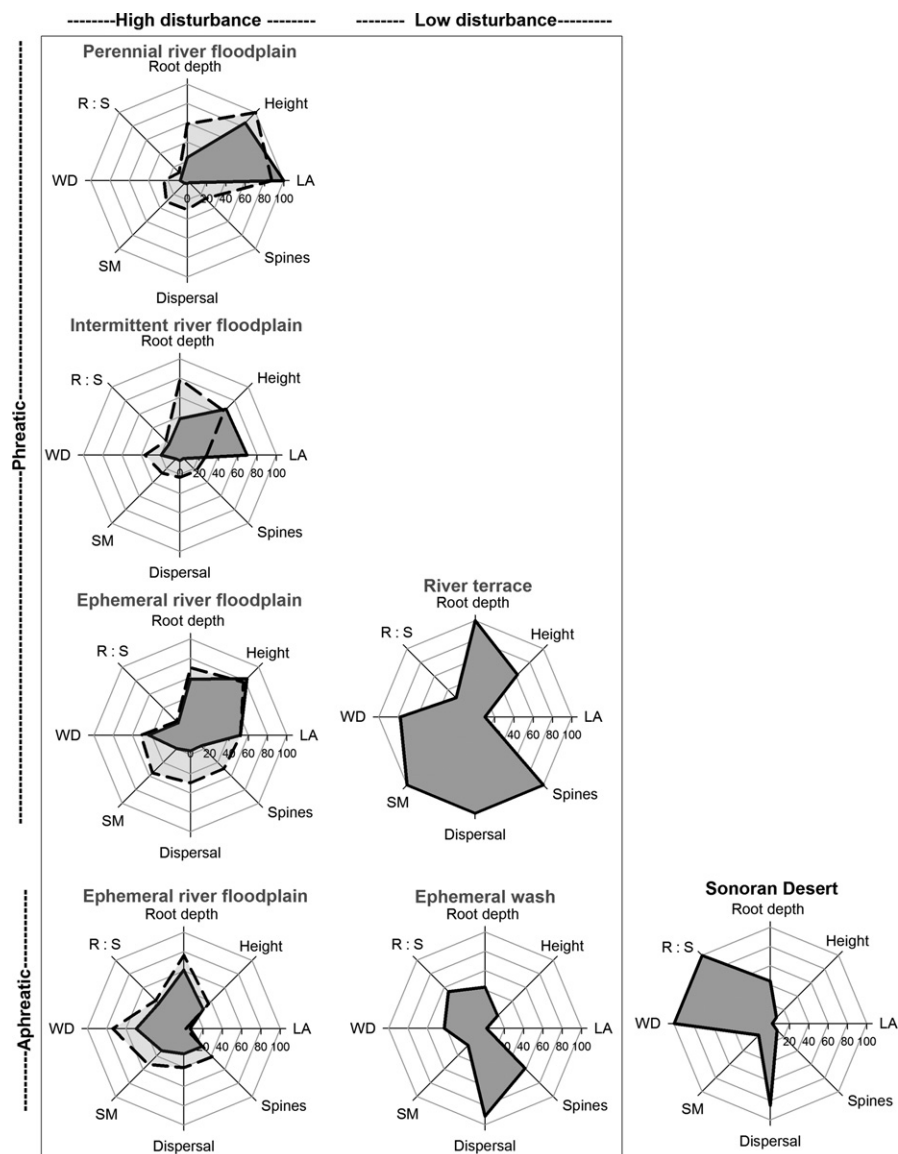


Fig. 6 Radar plots showing community-weighted traits for riparian habitat types that vary in surface and subsurface water flows and in fluvial disturbance. For the plots on the far left, solid lines depict low floodplain surfaces and dashed lines depict high floodplain surfaces. Habitats in the bottom row do not have access to a water table. Traits are standardised to have maximum value of 100. Height, canopy height; LA, leaf area; Spines, spinescence; SM, seed mass; WD, wood density and R : S, root-to-shoot ratio). A high value for dispersal indicates animal (versus wind) dispersal; a high value for spinescence indicates presence of spines.

(Karrenberg, Edwards & Kollmann, 2002; Parolin, 2003; Merritt *et al.*, 2010).

Our hypothesis regarding reproductive strategies was supported in that small, wind- and water-dispersed seeds were more prevalent in areas of higher fluvial disturbance while animal-dispersed species prevailed in more stable habitats. The disturbance indicators used in this study, seed mass and dispersal morphology, responded as expected to indicators of fluvial disturbance (elevation above thalweg and catchment size) and helped to differentiate pioneer guilds affiliated with river floodplains from non-pioneer types affiliated with river terraces and small catchments (Lavorel *et al.*, 1997). Both traits reflect processes operating at the regeneration life stage, with small wind- and water-dispersed seeds enabling plants to quickly colonise newly disturbed sites

(Streng, Glitzenstein & Harcombe, 1989). Similar shifts from small to large seed size (Kyle & Leishman, 2009; Wilfahrt, Collins & White, 2014) and from wind- to animal-dispersed strategies (Drezner, Fall & Stromberg, 2001; Lamb & Mallik, 2003) have been reported along disturbance or successional gradients in other riparian (and upland) ecosystems.

The combination of high fluvial disturbance and extreme drought stress that occurs in many desert rivers produces strong adaptive challenges. Plant traits often vary along multiple environmental gradients (Westoby *et al.*, 2002), and many of the traits we examined covaried with water availability and fluvial disturbance. Of note, the finding that community-weighted seed mass was related to both fluvial disturbance and water availability (depth to water table) is consistent with prior

research suggesting that large seed size increases seedling survivorship in undisturbed (shaded) as well as dry habitats (Cornelissen, 1999; Stromberg & Boudell, 2013).

Water acquisition traits, acquisitive (fast) guilds and conservative (slow) guilds

Several traits, including leaf area, woody plant canopy height and wood density, distinguish between plants with fast growth, high productivity and rapid acquisition of resources (acquisitive type) and those with slow growth and high investment in storage organs or herbivore defence (conservative type) (Grime *et al.*, 1997; Westoby & Wright, 2006; Reich, 2014). These traits vary regionally along gradients of precipitation and aridity (Givnish, 1987; Scholes *et al.*, 2002). For the specialised habitat of riparian zones in a hot, arid desert region, these traits were tightly coupled to water availability (depth to ground water in particular). Tall stature, shallow roots, large leaves, porous (low density) wood and thus high growth rate (Grady *et al.*, 2013) typified areas with shallow water tables and ample seasonal surface flows. ('Large' leaf size needs to be placed into context, however, as even the largest-leaved guild in the hot desert riparian zones of this study falls within the mid-size mesophyll group of Raunkiaer's, 1934 global classification system.)

Where water is abundant, fast-growing, large-leaved, productive hydriparian guilds dominate. However, these guilds (e.g. *Populus fremontii* and *Salix gooddingii*) are vulnerable to drought-induced cavitation and, as water availability declines, are replaced by slower-growing, drought-resistant guilds with higher wood density and greater water use efficiency (Tyree, Davis & Cochard, 1994; Pockman & Sperry, 2000; Hacke *et al.*, 2009). The xeriparian (conservative) groups occurred in areas with deeper water tables and ephemeral flow, and were typified by small leaves and higher wood density.

The conservative groups were further differentiated by rooting depth. Root architecture determines a plant's ability to use water from different depths and sources, influences capacity to survive drought and scouring events and influences productivity (Schenk & Jackson, 2002; Karrenberg *et al.*, 2003; West *et al.*, 2012). The conservative guilds with deep roots (and thus access to deep riparian water tables) were able to produce tall canopies. Rooting depth has aided in discrimination among functional types in riparian (Liu *et al.*, 2005) as well as non-riparian habitats (Garcia-Mora, Gallego-Fernandez & Garcia-Novo, 1999; Pohl *et al.*, 2011),

although its use is limited by a lack of data (Canadell *et al.*, 1996; Stromberg, 2013).

Many of the traits that we expected to be linked exclusively with water availability were also related to fluvial disturbance. Canopy height varied strongly with water availability, but short canopy (thus less time to grow to maturity) also is selectively advantageous under very high disturbance (Westoby, 1998; Pakeman & Eastwood, 2013), explaining our finding of a positive relationship between canopy height and elevation above thalweg. Low-density wood allows for rapid growth to reproductive age given ample water and thus is adaptive in high resource environments that are periodically 'reset' by scouring floods. Conversely, high wood density is associated with highly hydrologically variable and unpredictable environments (Lawson, Fryirs & Leishman, 2015). Leaf length, too, was linked with water availability and fluvial disturbance. Although leaves perform a variety of gas and heat exchange functions, leaf morphology also reflects selective pressures that minimise tissue damage from mechanical stresses such as shear stress due to river flooding (Nicotra *et al.*, 2011). Linear structure (high leaf length: width ratio) may enhance hydrodynamics and reduce leaf tissue loss from flow-related scour (high shear stress), consistent with our finding that the linkage with fluvial disturbance was stronger for leaf length than it was for leaf area.

Many plants of arid environments produce spines to protect against foraging by large herbivores (Wana & Beierkuhnlein, 2011; Frenette-Dussault *et al.*, 2012). In our study system, spines were present in approximately half of the xeriparian and desert guilds, all associated with relatively stable habitats (river terraces, high floodplains, small desert washes, upland desert). In habitats that were dry and flood-scoured, notably floodplains of ephemeral rivers, plants did not allocate resources to mechanical defence, presumably because of the greater likelihood of dying from flood scour than from herbivory in such settings.

Future research

This study addressed guild structure of woody plants along ephemeral, intermittent and perennial rivers. To be more comprehensive, herbaceous flora should be included (Lavorel *et al.*, 1997; Willby, Abernethy & Demars, 2000; Zelnik & Carni, 2008). Herbaceous plant species outnumber woody species in riparian zones of dryland regions (Viers *et al.*, 2012; Stromberg & Boudell, 2013), particularly along intermittent streams where they are not limited by dense tree canopy (Stromberg *et al.*,

2009a,b; Katz, Denslow & Stromberg, 2012). Including herbaceous plants also is advisable because they respond to environmental change over shorter time-scales and are more sensitive to subtle changes in hydrology.

Although our study spanned wide ranges of water availability and fluvial disturbance, we did not address adaptive traits and functional types of wetlands that experience prolonged inundation but little fluvial disturbance (Kozlowski, 2002; Catford & Jansson, 2014). Such spring-fed wetlands have become rare in many arid regions (Minckley, Turner & Weinstein, 2013; Silcock, Healy & Fensham, 2015). Other stream types in need of greater study are those with ephemeral and intermittent surface flows, and those in mountainous headwaters and urban areas disconnected from ground water (Bate-man *et al.*, 2015).

Conservation science must be rooted in theory but be sufficiently practical for use. Additional traits may be needed to differentiate among functional types in broader regional assessments, but to maximise efficiency, we should search for a small number of non-redundant traits (Box, 1995; Laughlin, 2014). C : N ratio in leaves is potentially useful for differentiating plants that are chemically well-defended against ungulate grazing (a common land use in dryland regions) from those that are more palatable (Pringle *et al.*, 2011). Reproductive effort may help to separate abundant seed producers such as *T. chinensis* from other members of the xeroriparian guild. Capacity for resprouting should be summarised in a rigorous fashion (e.g. meristem density), as it likely differentiates among pioneer types. Finally, because traits and their relationship to the environment differ at different life stages, regenerative and adult guilds could be defined.

Another avenue of research is the degree of intraspecific variability in traits. For example, male and female *Acer negundo* differ in several physiological traits influencing water use efficiency (Dawson & Ehleringer, 1993). *Populus* may have denser wood, more cavitation-resistant xylem and slower growth if established in drier environments and may be less vulnerable to abrupt groundwater decline than individuals with lower wood density. This plasticity varies by species and influences hydrologic niche breadth (Stromberg, Wilkins & Tress, 1993; Scott, Lines & Auble, 2000; Shafroth, Stromberg & Patten, 2000) and could be included in the delineation of guilds.

Ultimately, riparian plant functional types can be linked with ecosystem services (Jones *et al.*, 2010; Bagstad *et al.*, 2012). Although this study focussed on response traits rather than effect traits, the categories

sometimes overlap (Suding *et al.*, 2008). Canopy height and leaf area, for example, are linked with services ranging from carbon acquisition (Lavorel & Garnier, 2002) to provision of bird habitat (Merritt & Bateman, 2012). Abundance of plants with edible seeds or fruits is important in sustaining particular guilds of mammals and birds (Koniak, Noy-Meir & Perevolotsky, 2009). To address concerns regarding declines in pollinators and crop-pollinating ecosystem services, future assessments could include morphological adaptations for pollen vectors (Sacchi & Price, 1988; Lamb & Mallik, 2003; Pake-man & Eastwood, 2013).

We view quantitative trait analysis and functional typing as complementary approaches to classic plant ecological studies that can provide insights into evolutionary biology, functional ecology and phylogenetic structure (McGill *et al.*, 2009). Gradient analyses that model quantitative traits as a function of driving variables that can be managed for (e.g. flow regimes) are increasingly important as they enable us to make predictions about the outcomes of possible futures, including shifts towards greater flow intermittency. Such predictions can inform decision-makers, influence policy and affect the magnitude and direction of future water development. Further, the development of guilds and quantifications of their functional and response characteristics can help us to set functional goals for management and restoration of rivers, be they ephemeral, intermittent or perennial.

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